Fiscal Year:	FY 2023	**Task Last Updated:**	FY 01/15/2024
PI Name:	Chancellor, Jeffery Ph.D.		
Project Title:	Integration of In-Silico and In-Vivo Models for Determining Preclinical Indicators and/or Integrated Biomarkers of Radiation-Induced Vascular Dysfunction		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019-2020 HERO 80JSC019N0001-HHCBPSR, OMNIBUS2: Human Health Countermeasures, Behavioral Performance, and Space Radiation-Appendix C; Omnibus2-Appendix D
Start Date:	06/01/2021	**End Date:**	05/31/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	5	**No. of Master' Degrees:**	2
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
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Performance Goal No.:			
Performance Goal Text:			

Task Description:	Background: The intravehicular (IVA) radiation environment onboard future NASA exploration space vehicles results in nonhomogeneous dose deposition of critical organ systems in humans. The research team has recently shown that the IVA radiation environment, as measured onboard the International Space Station (ISS), can be reproduced at terrestrial accelerator laboratories by impinging a monoenergetic beam of ions of a single species upon a passive scattering apparatus, which emits polyenergetic spectra of multiple ion species that emulate IVA radiation fields. This approach will enable us to perform systematic ground-based research studies to determine dose response and relative biological effectiveness (RBE) values of radiation-induced degenerative effects and explore the key mechanisms leading to radiation-induced accelerated aging. In small animal models, we will vary exposure factors, such as dose and dose rate, as well as key host factors (e.g., sex and age) of relevance to humans. We will also utilize radiation exposures that are physiologically scalable, allowing for mammalian models with smaller body mass to be irradiated with a nonhomogeneous heavy-ion field that physiologically replicates exposures in humans. Microfluidic human organ-on-chip systems will then be used to facilitate the translation of molecular markers from pre-clinical models to humans. Significance: This work will allude to possible precursor markers and/or integrated biomarkers of an underlying cardiovascular risk that is attributed to space flight radiation exposure and suggest possible biologic targets and pathways of relevance to novel countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Computational human phantoms are widely used for Monte Carlo dose calculations, however, their efficacy in predicting outcomes are limited because voxel-based human phantoms constructed using tomographic images of human physiology have several limitations. These include stair-stepped surfaces, finite voxel resolution that results in inaccurate dose calculations and limited ability to fully capture the anatomic complexities of human internal anatomy. The approach developed here will facilitate accurate 4D Monte Carlo dose topology calculations in mouse and human physiology using tetrahedral-mesh geometry based on human CT and MR data. This combined with direct transportation radiation particles will allow for modeling dynamic changes to brain structure by interpolation of deformation vector fields. Additionally, the results further demonstrate the utility of high-performance, multi-core computers and sophisticated numerical techniques to study complex interaction dynamics that are otherwise difficult to replicate in a laboratory setting. Allowing for incremental exploration of physical and biological phenomena in advance of expensive or prohibited laboratory experiments.
	The current specific aims as currently funded are as follows: Specific Aim 1: Replicate the relative organ dose distribution seen in humans exposed to radiation in deep space vehicles in an appropriate mammalian model(s). Specific Aim 2: Identify functional and cellular indicators of cardiovascular function in a mouse model of space-relevant radiation exposures. Specific Aim 3: Determine which cardiovascular and gastrointestinal (GI) outcomes of simulated GCR exposure in the mouse translate to human organs on chip. Specific Aim 4a: Demonstrate and validate that the IVA radiation environment, as measured onboard the ISS, can be reproduced at terrestrial accelerator laboratories by impinging a monoenergetic beam of ions of a single species upon a passive scattering apparatus, which emits poly-energetic spectra of multiple ion species that emulate IVA radiation fields. Specific Aim 4b: Using the SimGCRSim, identify functional and cellular indicators of cardiovascular function in a mouse model of space-relevant radiation exposures and compare to those found in Specific Aim 2. Progress to date on Specific Aim 1: We simulated the intravehicular radiation environment onboard the International Space Station (ISS) and the corresponding exposure in the MOuse Whole BodY (MOBY) phantom using 3D Monte Carlo particle transport. We then modeled the radiation spectra from the NASA Space Radiation Laboratory Simple Galactic Cosmic Ray Simulator and determined the dose distribution within the MOBY. We tallied absorbed dose in organs and issues throughout the whole body for both simulations. Absorbed dose data from the mouse exposure was compared with corresponding data from a companion study of intrahevicular exposure to a human phantom. Deviation in the dose falloff within the MOBY phantom was found for both simulations compared to the human phantoms. The methods for determining the organ dose to humans on ISS are described in a manuscript is in peer review with Nature Microgravity and noted in the bibliography for reference. Several physical beamline setups were then computationally configured, and the percent deviation from the human phantom values was used as training data for a machine-learning evolutionary algorithm based on Darwin's theory of evolution. An evolutionary algorithm is a good choice for this problem as they excel in mul-objective optimization. The geometry calculated by the genetic algorithm predicts an average organ dose percent deviation of 5.16% and a max organ dose percent deviation of 10.94%, as demonstrated in Table 1. These preliminary results indicate that the relative organ dose in a mouse model following current space radiation exposure methods does not correspond to doses predicted for human astronauts in Low Earth Orbit, indicating the need for alterations to radiation environments in murine studies. Progress to date on Specific Aim 2: SA2 is currently in progress and will reported following the completion of SA4a, SA4b. Preliminary results indicate subtle changes to cardiac health in the cohorts exposed to 0.5 Gy (@ 8hrs). A full presentation of the results will be presented the HRP Investigators workshop in February 2024. Progress to date on Specific Aim 3: Specific aim 3 will be performed during year 4 of the grant following the completion of SA2 and SA4b Progress to date on Specific Aim 4a: For simplicity, we focus on the IVA radiation spectrum measured on the ISS. Computational methods for developing the geometry are described in the original grant proposal. The target moderator block is constructed from materials with multiple layers of varying geometry to generate specific nuclear reactions and spallation products using 3D printing technology. This target moderator block will be placed in front of a 1 GeV electron volt per nucleon (GeV/n) iron (56Fe) single- particle beam with no modifications to the beamline infrastructure. As the iron beam passes through the moderator block, nuclear spallation processes can create modest amounts of the desired fragments resulting in a complex mixed field of particle nuclei with different atomic numbers Z in the range $0 < Z = 26$ and LETs up to approximately 200 keV/µm. Detectors downstream of the target block will measure the field to characterize the number of surviving primary ions (56Fe). These measurements will be compared to those predicted in the computational development. Analysis will be done immediately following each measurement, and if needed, real-time incremental changes to the target block geometry will be done on site using 3D printers. Measurements were

Task Progress:	taken using both the 20x20cm beam, and the 60x60cm beam during the Spring 2023 and Fall 2024 NSRL beam campaign. Scattering aperture was constructed using build-out, liquid resin 3-D printers were used to fabricate the scaUering apertures. These printers have resolu4ons of 30 microns in the XY plane and 10 microns in the Z plane along the build axis. Each channel has four apertures with a surface area of 1cm x 1cm in the XY plane, so the field generated would be homogenous across the exposure plane. Additionally, each aperture was 22 cm in length along the Z-plane. A total of 900 were printed, and 9 20 cm x 20 cm blocks were constructed for a total area of 60 cm x 60 cm. The total fabrication cost is approximately $6,000 and takes about 40 days to complete Beamline measurements of the field produced by the moderator block were taken at NSRL in April 2023. Two detector tunnels were used. The first is an array of five dEdx plastic scintillators placed downstream of the block. Two were placed 10 cm behind the aperture; two were placed 1.5 meters behind the aperture. A plastic scintillator calorimeter also followed these two. This facilitated five discreet energy deposition for determining the LET and particle species spectra and the amount of surviving primary ions (510 MeV/n iron). Determining the LET and particle spectra are in process, with anticipated completion Spring 2024. The relative number of primary ions surviving is within a 3% margin of error in the computational predicted value. A significant finding that precludes the field is being accurately produced. Additionally, twelve semiconductor Timepix detectors were used for measuring field species and neutron production. The list of detectors and their arrangement at the beamline is enclosed at the end of this document. The complete analysis of the measurements taken is ongoing and conducted by two of my graduate students at LSU and one graduate student in Prague, Cz., at the Advacam laboratory. Results will be reported as completed, but the total effort is anticipated to be about one year. Photos of the detector arrays are enclosed here. Progress to date on Specific Aim 4b: At the request of the NASA Program Officer, we used the NSRL simple GCRsim on the first cohort of animals during the Spring 2022 campaign, each with 10 male and 10 female animals: (1) 0.5 Gy + HLU (7.92 hours), (2) 1.5 Gy + HLU (24 hours), (3) 0 Gy + HLU, (4) 0 Gy, (5) 0.5 Gy. full-weight bearing. After exposure, the animals were returned to UAMS. where gastrointestinal (GI), in vivo cardiac and vascular function, pathological changes in the heart and vasculature, as well as plasma markers of inflammation and cardiovascular disease were measured at 2 weeks and 9 months after the completion of irradiation. The exposure of 96 age-appropriate male and female C57BL/6J mice to doses 500 mGy and 1.5 Gy was planned and partially executed during the Spring 2023 NSRL beam campaign. Exposure times would be 7.92 hrs, and 24 hrs, respectively. Mice will be subjected simultaneously to hind-limb unloading (HU) to model fluid shifts in microgravity. An additional cohort of both male and female non-HU mice will be exposed to 0 Gy and 500 mGy. However, because of problems with the beamline vacuum, NSRL was unable to provide sufficient beam intensity for the proposed studies. In response, the LSU and Wake Forest team worked with NSRL, NASA SR to transfer the animals to a Duke University team funded by NASA Space Biology. A cohort of full weight bearing animals was exposed to a dose of 1.5 Gy over a 24-hr time period using the NSRL 5-ion GCRSim. At the request of the NASA Program Officer, we used the NSRL 28-ion GCRsim on the first cohort of animals during the Fall 2023 campaign, each with 10 male and 10 female animals: (1) 0.5 Gy + HLU (7.92 hours), (2) 1.5 Gy + HLU (24 hours), (3) 0 Gy + HLU, (4) 0 Gy, (5) 0.5 Gy full-weight bearing., and (6) 1.5 Gy "acute exposure" given over a 4 hr period. After exposure, the animals were returned to UAMS. where gastrointestinal (GI), in vivo cardiac and vascular function, pathological changes in the heart and vasculature, as well as plasma markers of inflammation and cardiovascular disease were measured at 2 weeks and 9 months after the completion of irradiation. These results will be compared to the simple GCRSim and reported late 2024 or early 2025.

Bibliography Type:	Description: (Last Updated: 02/19/2024)
Articles in Other Journals or Periodicals	Chesal MA, Blue RS, Aunon-Chancellor SA, Chancellor JC. "Novel tetrahedral human phantoms for space radiation dose assessment." arXiv preprint server. Posted August 23, 2023. https://doi.org/10.48550/arXiv.2303.05564 , Aug-2023
Articles in Peer-reviewed Journals	Chancellor JC. "The impact of ions on DNA." Physics. 2023 Mar 13;16:41. http://dx.doi.org/10.1103/Physics.16.41 , Mar-2023
Articles in Peer-reviewed Journals	McMahon MD, Chancellor JC. "Accident simulation study for nuclear power plants impacting Louisiana: differences in 2017 vs. 1992 protective action guidelines." Health Phys. 2022 Sep 1;123(3):218-28. http://dx.doi.org/10.1097/HP.0000000000001589 ; PMID: 35678725 , Sep-2022
Articles in Peer-reviewed Journals	Afshari N, Koturbash I, Boerma M, Newhauser W, Kratz M, Willey J, Williams J, Chancellor J. "A review of numerical models of radiation injury and repair considering subcellular targets and the extracellular microenvironment." Int J Mol Sci. 2024 Jan 13;25(2):1015. Review. https://doi.org/10.3390/ijms25021015 ; PMID: 38256089; PMCID: PMC10816679 , Jan-2024
Awards	Chesal M. "LaSPACE Graduate Research Fellowship, July 2021." Jul-2021
Awards	Miller J. "LaSPACE Undergraduate Research Assistantship, July 2021." Jul-2021
Awards	Wilkie D. "LaSPACE Undergraduate Research Assistantship, August 2021." Aug-2021
Awards	Chesal M. "LaSPACE Graduate Research Fellowship, August 2021." Aug-2021
Awards	Fields G. "LaSPACE Undergraduate Research Assistantship, August 2023." Aug-2023
Awards	Afshari N. "LaSPACE Graduate Research Fellowship, August 2023." Aug-2023
Awards	Manning S. "LaSPACE Graduate Research Fellowship, August 2023." Aug-2023
Awards	Miller J. "Fulbright Award, Study/Research Grant (2022-2023), May 2022. " May-2022

Awards	Nemec-Bakk A. "Translational Research Institute for Space Health (TRISH) Postdoctoral Award, July 2022." Jul-2022
Dissertations and Theses	Taylor J. "An artificial intelligence framework to design radiation shielding for spacecraft and satellites using machine learning and topology optimization." Dissertation, Louisiana State University, May 2023. , May-2023
Dissertations and Theses	Chesal M. "Computational mammalian phantoms for predicting dose topology incurred in the space radiation environment." Masters Thesis, Louisiana State University, May 2022. , May-2022
Dissertations and Theses	McMahon M. "Accident simulation study for nuclear power plants impacting Louisiana." Masters Thesis, Louisiana State University, August 2022. , Aug-2022
Patents	63/257,434. Provisional patent issued June 2021. Jun-2021 Chancellor JC, Chesal MA. "System and Method for Simulating Non-Homogeneous Space Radiation Environment."